# Samira Arfaee

*Ph.D. Candidate in Mathematics*

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## Research Interests

Probability and algebraic combinatorics, with particular interest in Markov chains and mixing times on the symmetric group and its type A Iwahori–Hecke algebra, using representation-theoretic and algebraic methods.

## Education

2021–Present **Ph.D. in Mathematics**, *Stony Brook University*, Stony Brook, NY, USA
Advisor: **Evita Nestoridi**.

2019–2021 **M.Sc. in Mathematics**, *École Polytechnique Fédérale de Lausanne (EPFL)*, Lausanne, Switzerland

2014–2019 **B.S. in Mathematics and Applications**, *Amirkabir University of Technology (AUT)*, Tehran, Iran
Second rank.

2016–2019 **Minor in Electrical Engineering (Communications)**, *Amirkabir University of Technology (AUT)*, Tehran, Iran

## Publications and Preprints

2026 *Cutoff for q-deformed classical card shuffles in the type A Iwahori–Hecke algebra*, with Bryan Wong. arxiv:2609.13570.

2025 *Shuffling via sums of Jucys–Murphy Elements*, with Evita Nestoridi. arXiv:2504.07918.

2023 *A new approach to character-free proof for Frobenius theorem*, with Vahid Shahverdi. *AUT Journal of Mathematics and Computing*, Vol. 4, No. 1, pp. 99–103.

## Honors and Awards

2026 **Chairman's Award for Excellence in Graduate Studies( For exceptional contributions to research, teaching, and service.)**, *Stony Brook University*

2018 **Second Prize**, *International Mathematics Competition (IMC)*, Blagoevgrad, Bulgaria

2018 **Third Prize**, *Al-Khorezmi International Mathematical Olympiad (AKIMO)*, Uzbekistan

2017 **Silver Medal**, *Iranian Mathematics Competition for University Students*

2016 **Bronze Medal**, *Iranian Mathematics Competition for University Students*

## Professional Service

Refereeing Referee for *The Annals of Applied Probability* and *SIAM Undergraduate Research Online (SIURO)*.

## Invited Talks and Presentations

Nov. 2026 **CAGE (Combinatorics, Algebra and Geometry) Seminar**, *University of Pennsylvania*, Philadelphia, PA, USA

Jul. 2026 **4th Simons Math Summer Workshop: Algebraic Methods in Probability**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

Jul. 2026 **4th Simons Math Summer Workshop: Algebraic Methods in Probability**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

Apr. 2026 **AMS Spring Central Sectional Meeting**, *North Dakota State University*, Fargo, ND, USA

Oct. 2025 **AMS Fall Southeastern Sectional Meeting**, *Tulane University*, New Orleans, LA, USA

Mar. 2025 **Recent Developments on Mixing Times**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

## Conferences, Workshops, and Summer Schools

Dec. 2025 **Webs in Algebra, Geometry, Topology, and Combinatorics**, *ICERM*, Providence, RI, USA

Jun-Jul 2025 **Graphical Models in Algebraic Combinatorics**, *Summer Graduate School, SLMath*, Berkeley, CA, USA

Feb. 2025 **Connections Workshop: Extremal Combinatorics**, *SLMath*, Berkeley, CA, USA

Jan. 2025 **Introductory Workshop: Probability and Statistics of Discrete Structures**, *SLMath*, Berkeley, CA, USA

Jan. 2025 **Connections Workshop: Probability and Statistics of Discrete Structures**, *SLMath*, Berkeley, CA, USA

Jul. 2024 **Virginia Integrable Probability Summer School**, *University of Virginia*, Charlottesville, VA, USA

Jun. 2024 **The Many Combinatorial Legacies of Richard P. Stanley**, *Harvard University*, Cambridge, MA, USA

Mar. 2024 **Universality & Integrability in KPZ**, *Columbia University*, New York, NY, USA

Oct. 2023 **The Asymmetric Simple Exclusion Process**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

## Teaching Experience

### Stony Brook University

Summer 2025, 2026 **Lecturer**, *Simons STEM Scholars Program*, Stony Brook University
Invited to return in Summer 2026 after serving in Summer 2025.

Fall 2025 **Lecturer**, *Trigonometry and Logarithms*, Stony Brook University
Two sections.

Spring, Fall 2024 **Lecturer**, *Proficiency Algebra*, Stony Brook University
Substitute instructor in Spring 2024.

Fall 2023; Summer 2024 **Lecturer / Recitation Instructor**, *Calculus I*, Stony Brook University
Recitation Instructor, two sections, Fall 2023; Lecturer, Summer 2024.

Summer 2022, Spring 2023, Summer 2023 **Lecturer / Recitation Instructor**, *Calculus A*, Stony Brook University
Lecturer, Summer 2022 and Summer 2023; Recitation Instructor, two sections, Spring 2023.

Fall 2022 **Recitation Instructor**, *Linear Algebra*, Stony Brook University
Two sections.

Summer 2022 **Lecturer**, *Calculus III and Applications*, Stony Brook University

Fall 2021 **Recitation Instructor**, *Calculus B*, Stony Brook University
Two sections.

### Earlier Teaching

Fall 2020 **Teaching Assistant**, *Discrete Mathematics*, EPFL, Lausanne, Switzerland

Fall 2017 **Teaching Assistant**, *Foundations of Mathematical Analysis*, Amirkabir University of Technology, Tehran, Iran

Spring, Fall 2017 **Teaching Assistant**, *Foundations of Matrix and Linear Algebra*, Amirkabir University of Technology, Tehran, Iran

Spring 2017 **Teaching Assistant**, *Foundations of Algebra*, Amirkabir University of Technology, Tehran, Iran

## Mentoring

Fall 2025 **Mentor**, *Directed Reading Program*, Stony Brook University
Mentored undergraduate students in probability, group theory, and Markov chains; guided independent reading, proof-writing, and presentations.